COMPASS: Convex Optimization of Mediated Polygenic Architecture via SNP Scores

Abstract

We present a unified methodological framework for estimating gene-specific mediated heritability from genome-wide association study (GWAS) summary statistics using gene-variant annotations. While Mediated Expression Score Regression (MESC) relies on observed eQTL effect sizes, COMPASS can leverage arbitrary gene-variant annotations (e.g., deep learning predictions, Hi-C contact frequencies). We introduce a convex optimization framework that identifies modular, multi-mechanism regulatory architectures across disease genes. We derive an alternating projection algorithm combining Singular Value Thresholding (SVT) and non-negative clipping to fit the genome-wide model efficiently.

1 Model

We start with the same setup as MESC. Let ω_j denote the marginal effect of single nucleotide polymorphism (SNP) $j \in \{1, \ldots, J\}$ on a complex trait. We model ω_j as a causal cascade mediated through genes $k \in \{1, \ldots, K\}$:

$$\omega_j = \sum_{k=1}^{K} \beta_{jk} \alpha_k + \gamma_j \tag{1}$$

where β_{jk} is the true, unobserved regulatory effect of SNP j on gene k, α_k is the causal effect of gene k on the trait, and $\gamma_j \sim \mathcal{N}(0, \tau_\gamma)$ captures non-mediated (direct or pleiotropic) effects.

Instead of relying on empirical eQTLs, we utilize L non-negative annotations $a_{jkl} \geq 0$ representing the predicted regulatory potential of SNP j on gene k via biological mechanism $l \in \{1, \ldots, L\}$. We place a linear variance prior on the regulatory effects:

$$\beta_{jk} \sim \mathcal{N}(0, V_{jk}) \quad \text{where} \quad V_{jk} = \sum_{l=1}^{L} w_l a_{jkl} \tag{2}$$

Here, $w_l \geq 0$ represents the unknown heritability contribution weight of mechanism l.

1.1 The infinitesimal Gaussian assumption and its limitations

A simple assumption would be infinitesmal effects, $\alpha_k \sim \mathcal{N}(0, \tau_\alpha)$. Given that β_{jk} and α_k are independent and mean-zero, the variance of their product is the product of their variances:

$$\mathrm{Var}(\beta_{jk}\alpha_k) = \mathbb{E}[\beta_{jk}^2]\mathbb{E}[\alpha_k^2] = V_{jk}\tau_\alpha \tag{3}$$

Assuming independence across genes, we sum over genes, substitute the linear variance model for V_{jk}, and commute the sums:

$$\mathrm{Var}(\omega_j) = \sum_{k=1}^{K} V_{jk}\tau_\alpha + \tau_\gamma = \tau_\alpha \sum_{k=1}^{K}\left(\sum_{l=1}^{L} w_l a_{jkl}\right) + \tau_\gamma = \tau_\alpha \sum_{l=1}^{L} w_l \left(\sum_{k=1}^{K} a_{jkl}\right) + \tau_\gamma = \sum_{l=1}^{L}(\tau_\alpha w_l) A_{jl} + \tau_\gamma, \tag{4}$$

where $A_{jl} \equiv \sum_{k=1}^{K} a_{jkl}$. The gene-aggregated matrix A is therefore simply a $J \times L$ SNP annotation matrix: each column $A_{\cdot l}$ can be used as an ordinary S-LDSC annotation with coefficient $\theta_l = \tau_\alpha w_l$. Thus this model is almost trivial, involving just summing over genes.

1.2 Gene-specific resolution via conditional variance

To achieve gene-specific resolution we cannot rely on the infinitesimal Gaussian assumptions above. Here we consider conditioning the variance of the marginal effect on the specific causal trait variance of each gene, $s_k = \alpha_k^2 \geq 0$,

$$\mathrm{Var}(\omega_j \mid \alpha) = \sum_{k=1}^{K} s_k V_{jk} + \tau_\gamma \tag{5}$$

Substituting this conditional variance into the expected GWAS χ^2 statistic yields,

$$\begin{aligned}
\mathbb{E}[\chi_j^2 \mid \alpha] &= 1 + N \sum_{m=1}^{J} r_{jm}^2 \mathrm{Var}(\omega_m \mid \alpha) \\
&= 1 + N \sum_{m=1}^{J} r_{jm}^2 \left(\sum_{k=1}^{K} s_k \sum_{l=1}^{L} w_l a_{mkl} + \tau_\gamma\right) \\
&= 1 + N \sum_{k=1}^{K}\sum_{l=1}^{L}(s_k w_l)\mathcal{T}_{jkl} + N\tau_\gamma l_j
\end{aligned} \tag{6}$$

where r_{jm} is LD, $l_j = \sum_m r_{jm}^2$ is the standard non-mediated LD score, and the deterministic, pre-computable, sparse 3D annotation-gene LD Tensor,

$$\mathcal{T}_{jkl} = \sum_{m=1}^{J} r_{jm}^2 a_{mkl}. \tag{7}$$

Direct optimization of s and w has two pathologies:

1. **Scale Non-Identifiability:** For any scalar $c > 0$, the transformation $s \to s/c$ and $w \to cw$ leaves the product $B = sw^T$ unchanged, rendering the individual scales of gene importance and annotation importance unidentifiable without arbitrary anchor priors. This is addressable, for example, by constraining the L_1 norm of w (or s) to be 1, but this does not resolve the second pathology:

2. **Non-Convexity:** The bilinear optimization is non-convex, containing saddle points and local minima where optimization routines can get trapped.

1.3 Convex Nuclear Norm Relaxation

Let $B_{kl} = s_k w_l$ define the $K \times L$ matrix of mediated heritability contributions. We relax the strict rank-1 constraint $B = sw^T$ by treating $B \in \mathbb{R}^{K\times L}$ as a general matrix parameter regularized by the nuclear norm $\|B\|_* = \sum_i \sigma_i(B)$, where $\sigma_i(B)$ are the singular values of B.

We formulate the inference as a Weighted Ordinary Least Squares (WOLS) regression minimized over the space of non-negative matrices:

$$\min_{B\in\mathbb{R}_+^{K\times L},\, \tau_\gamma\geq 0} \quad \sum_{j=1}^{J} \eta_j \left(\chi_j^2 - 1 - N \sum_{k=1}^{K}\sum_{l=1}^{L} B_{kl}\mathcal{T}_{jkl} - N\tau_\gamma l_j \right)^2 + \lambda\|B\|_* \tag{8}$$

where η_j are standard S-LDSC regression weights accounting for LD redundancy and heteroskedasticity.

This relaxation yields three (potential) advantages:

- **Convexity:** Because the WOLS loss is convex with respect to B and the nuclear norm is the convex envelope of the rank function, the objective is convex, guaranteeing that any local minimum is the unique global optimum.
- **Discovery of Regulatory Modules:** Rather than enforcing regulatory homogeneity (where every causal gene obeys the exact same regulatory weight vector w), a low-rank solution $\text{rank}(B) = r > 1$ decomposes into r distinct mechanistic archetypes via Singular Value Decomposition (SVD):
$$B = \sum_{i=1}^{r} \sigma_i u_i v_i^T \tag{9}$$
Here, the right singular vectors $v_i \in \mathbb{R}^L$ represent distinct biological pathways (e.g., promoter-driven vs. distal Hi-C loop enhancer-driven), while the left singular vectors $u_i \in \mathbb{R}^K$ cluster disease genes by their physical mechanism of action.
- **Computational Tractability:** Because the number of annotations L is much smaller than the number of genes ($K \approx 20,000$), computing the reduced SVD during optimization takes $O(KL^2)$ time.

2 Cell-Type Indexing

The index l need not correspond to a biochemical mechanism such as promoter activity, enhancer activity, or chromatin contact class. The same derivation applies if l indexes cell types or cellular contexts. In particular, consider the model

$$\beta_{jkl} \sim \mathcal{N}(0, w_l a_{jkl}) \tag{10}$$

where l indexes cell types. We now model,

$$\omega_j = \sum_{k=1}^{K}\sum_{l=1}^{L} \beta_{jkl}\alpha_{kl} + \gamma_j \tag{11}$$

where α_{kl} is the causal effect of gene k in cell type l on the trait. We then have,

$$\text{Var}(\omega_j|\alpha) = \sum_{k=1}^{K}\sum_{l=1}^{L} \alpha_{kl}^2 w_l a_{jkl} + \tau_\gamma = \sum_{k=1}^{K}\sum_{l=1}^{L} B_{kl} a_{jkl} + \tau_\gamma \tag{12}$$

where now $B_{kl} = \alpha_{kl}^2 w_l$ is the gene-by-cell-type mediated heritability matrix. The same convex nuclear norm relaxation applies, and the SVD of B now identifies cell-type-specific regulatory modules.

2.1 Hierarchical context and gene-deviation model

Applying the nuclear-norm penalty directly to the complete matrix B also penalizes a context-wide effect shared by all genes. This differs from the infinitesimal S-LDSC reduction, in which every gene contributing to annotation l shares one unpenalized coefficient. To separate these two levels, we write

$$B_{kl} = s\, q_l + D_{kl}, \qquad s \geq 0, \quad q_l \geq 0, \quad D_{kl} \geq 0, \tag{13}$$

where q is a context profile, s is its fitted genome-wide scale, and D contains gene-specific departures from that profile. The resulting objective is

$$\min_{s,\, D,\, \tau_\gamma \geq 0} \sum_j \eta_j \left[\chi_j^2 - 1 - N \sum_{k,l} (s q_l + D_{kl}) \mathcal{T}_{jkl} - N \tau_\gamma l_j \right]^2 + \lambda \|D\|_*. \tag{14}$$

Thus, large λ suppresses only gene-specific deviations and leaves the context-wide component estimable. When q is unconstrained, the model estimates one non-negative coefficient per context. For the BaselineLD-calibrated analysis, q is the vector of official S-LDSC coefficients after setting negative entries to zero; COMPASS estimates s rather than transferring the reference-specific coefficient magnitude. Chromosome jackknife estimates are used for uncertainty in sq_l, with the S-LDSC profile uncertainty propagated approximately.

Jointly re-estimating s at every value of λ creates an avoidable allocation ambiguity: when the deviation penalty is weak, D can reproduce broad context signal and drive s to zero. We therefore use a sequential hierarchical estimator. Within each training fold, we first estimate s and τ_γ at the context-only end of the path, where $D = 0$. We then hold the resulting sq fixed while cross-validating λ for D; only D and τ_γ are updated. The final full-data fit follows the same procedure. Consequently, the shared context effect has one definition across the regularization path, and held-out improvements measure gene-specific signal conditional on that shared component rather than reallocating it.

The calibrated analysis is deliberately a two-stage conditional model. Because q is estimated from the same GWAS, its cell-type ordering is not an independent COMPASS discovery, and cross-validation selects only the penalty on D conditional on q. The analysis asks whether gene-specific departures improve held-out prediction while retaining a BaselineLD-adjusted context contrast; it is not an unbiased end-to-end validation of that contrast.

For interpretation, we report the context-wide and gene-specific contributions separately. If $c_l = \sum_j A_{jl}$ is the number of variants in the binary context annotation, the direct heritability attributed to context l is

$$H_l^{\text{global}} = c_l s q_l, \qquad H_l^{\text{deviation}} = \sum_{j,k} a_{jkl} D_{kl}, \qquad H_l^{\text{total}} = H_l^{\text{global}} + H_l^{\text{deviation}}. \tag{15}$$

This decomposition tests whether a weakly penalized deviation matrix changes the aggregate cell-type ranking even though the global component is fixed.

2.2 The Necessity of Non-Negativity

While standard summary statistic methods occasionally tolerate negative enrichment parameters, enforcing $B_{kl} \geq 0$ is mandatory in this gene-centric framework. Without non-negativity constraints, the optimization landscape suffers from two failures:

LD Collinearity Overfitting. Neighboring genes on a chromosome reside within the same LD block, causing their annotation tensors to be highly collinear ($\mathcal{T}_{j,k_1,l} \approx \mathcal{T}_{j,k_2,l}$ for adjacent genes k_1, k_2). If an unconstrained model attempts to fit a localized GWAS signal corresponding to a true heritability contribution of +10, it can overfit statistical noise by assigning extreme, mutually cancelling coefficients to adjacent genes, e.g. $B_{k_1,l} = +100$ and $B_{k_2,l} = -90$ so that $B_{k_1,l} + B_{k_2,l} = +10$. This "cancellation cheat" preserves the macroscopic model fit while destroying gene-level resolution, creating phantom causal effects. The constraint $B_{kl} \geq 0$ acts as a regularizer against LD collinearity, forcing the model to assign heritability exclusively to the most parsimonious gene.

Mathematical and Biological Plausibility By definition, $B_{kl} = \alpha_k^2 w_l$ represents the variance of the causal trait effect mediated by gene k via mechanism l. Variance cannot be negative. Further, because $\mathcal{T}_{jkl} \geq 0$, allowing negative entries in B permits the model to predict $\mathbb{E}[\chi_j^2] < 1$, implying that genetic variation suppresses sampling noise—a statistical impossibility under the null hypothesis.

2.3 Optimization via Alternating Projections

Because standard Singular Value Thresholding (SVT) introduces orthogonal vectors with negative values, we must solve a Non-Negative Low-Rank Matrix Approximation problem. We solve this efficiently using Proximal Gradient Descent with Dykstra's Alternating Projections.

Let $f(B, \tau_\gamma)$ denote the WOLS loss function from Equation (7). The algorithm proceeds as follows:

Algorithm 1 Non-Negative Low-Rank Gene Heritability Regression

Input: Pre-computed tensor $\mathcal{T} \in \mathbb{R}_+^{J \times K \times L}$, GWAS $\chi^2 \in \mathbb{R}^J$, LD scores $l \in \mathbb{R}^J$, step size ζ, regularization λ.
Initialize: $B^{(0)} \leftarrow \mathbf{0}_{K \times L}$, $\tau_\gamma^{(0)} \leftarrow 0$
while not converged **do**
    1. Gradient Step:
    $G^{(t)} \leftarrow B^{(t)} - \zeta \nabla_B f(B^{(t)}, \tau_\gamma^{(t)})$
    $\tau_\gamma^{(t+1)} \leftarrow \max\left(0,\ \tau_\gamma^{(t)} - \zeta \nabla_{\tau_\gamma} f(B^{(t)}, \tau_\gamma^{(t)})\right)$
    2. Singular Value Thresholding (Low-Rank Projection):
    Compute reduced SVD: $U, \Sigma, V^T = \mathrm{SVD}(G^{(t)})$
    $\tilde{\Sigma} \leftarrow \mathrm{diag}\left(\max(0,\ \sigma_i - \zeta\lambda)\right)$
    3. Non-Negative Projection:
    $B^{(t+1)} \leftarrow \max\left(\mathbf{0}_{K \times L},\ U\tilde{\Sigma}V^T\right)$
end while
Output: Modular heritability matrix $B^* = B^{(t+1)}$, background pleiotropy $\tau_\gamma^* = \tau_\gamma^{(t+1)}$.

This algorithm converges to the global optimal modular architecture, avoiding non-convexity and LD collinearity artifacts.

2.4 Cross-validation strategy

Leaving out an entire chromosome is not a valid penalty-selection scheme for determining λ. For example, if chromosome 1 is held out, then genes on chromosome 1 have no training annotations. Prediction on the held-out chromosome therefore relies on gene effects that were not estimable in the training set.

We therefore use the global, LD-aware procedure in Algorithm 2. The essential change is to define folds for genomic regions before considering genes. We construct a graph on variants, joining variants whose reference-panel LD exceeds a threshold $r_{jm}^2 \geq \rho_{\mathrm{CV}}$, and use its connected components as indivisible holdout units. Each component receives one genome-wide fold label, so the same fold has the same meaning for every gene and a variant linked to multiple genes cannot receive conflicting labels. Components are distributed across folds to balance the number of regression rows and, where possible, to place different components linked to the same gene in different folds.

Let $\mathcal{C}$ denote the set of CRE–gene links, and write $j \in c$ when variant j lies in CRE c. After the LD components have been assigned labels z_q, define

$$\begin{aligned}
\mathcal{K}(j) &= \{k : \exists\,(c,k) \in \mathcal{C} \text{ with } j \in c\},\\
\mathcal{K}_{-p} &= \{k : \exists\,(c,k) \in \mathcal{C},\ \exists\, q \text{ with } c \cap D_q \neq \varnothing \text{ and } z_q \neq p\},\\
\mathcal{S}_p &= \{j \in V_p : \mathcal{K}(j) \cap \mathcal{K}_{-p} \neq \varnothing\}.
\end{aligned} \tag{16}$$

Here, $\mathcal{K}(j)$ is the set of genes linked to variant j; $\mathcal{K}_{-p}$ is the set of "training" genes for fold p, i.e. that have at least one CRE overlapping a training LD component; and $\mathcal{S}_p$ is the set of held-out variants linked to at least one training gene. The intersection criterion allows a scored variant to have additional linked genes without training support. Those genes do not disqualify the variant: their effects remain unmodeled validation signal, making the score conservative rather than removing an otherwise informative row.

For fold p, all regression rows in V_p are removed from the fitting loss. When scoring $j \in \mathcal{S}_p$, prediction is restricted to the supported genes,

$$\widehat{\chi}^{2,(-p)}_{j,p\lambda} = 1 + N \sum_{k \in \mathcal{K}_{-p}} \sum_{l=1}^{L} \widehat{B}^{(p,\lambda)}_{kl} \mathcal{T}_{jkl} + N \widehat{\tau}^{(p,\lambda)}_{\gamma} l_j. \tag{17}$$

Variants outside CREs also inherit their component's fold and are held out of the training loss, preventing them from carrying correlated GWAS signal across the fold boundary, but they do not themselves contribute to the validation score used to choose λ.

Algorithm 2 Global LD-component cross-validation for selecting λ

Input: CRE–gene links; GWAS variants; A and R^2; descending penalty grid Λ; F folds; LD threshold ρ_{CV}
Construct $G = (V, E)$ with $V = \{1, \ldots, J\}$ and $E = \{(j, m) : r_{jm}^2 \geq \rho_{\text{CV}}\}$
Let $D_1, \ldots, D_Q$ be the connected components of G
Assign each D_q one global label $z_q \in \{1, \ldots, F\}$, balancing rows and spreading components linked to the same gene across folds
$\mathcal{P} \leftarrow \varnothing$
for $p = 1, \ldots, F$ **do**
  $V_p \leftarrow \bigcup_{q: z_q = p} D_q$ ▷ All rows in held-out LD components
  Compute $\mathcal{K}_{-p}$ and $\mathcal{S}_p$ from Equation 16
  if $\mathcal{S}_p = \varnothing$ **then**
    continue
  end if
  $\mathcal{P} \leftarrow \mathcal{P} \cup \{p\}$
  Set regression weight $\eta_j^{(p)} \leftarrow \eta_j \, \mathbf{1}[j \notin V_p]$
  Initialize $B \leftarrow 0$ and $\tau_\gamma \leftarrow 10^{-8}$
  for $\lambda \in \Lambda$ from largest to smallest **do**
    Fit (B, τ_γ) using weights $\eta^{(p)}$, warm-starting from the preceding λ
    Predict $\widehat{\chi}_{j,p\lambda}^{2,(-p)}$ using only genes in $\mathcal{K}_{-p}$
    $E_{p\lambda} \leftarrow |\mathcal{S}_p|^{-1} \sum_{j \in \mathcal{S}_p} (\chi_j^2 - \widehat{\chi}_{j,p\lambda}^{2,(-p)})^2$
  end for
end for
$\lambda^* \leftarrow \arg\min_{\lambda \in \Lambda} |\mathcal{P}|^{-1} \sum_{p \in \mathcal{P}} E_{p\lambda}$
Refit on all regression rows, again warm-starting from larger penalties down to λ^*
Output: λ^* and the full-data estimates $(\widehat{B}, \widehat{\tau}_\gamma)$

This design directly targets the desired generalization: the model learns B_{kl} from some regulatory regions for gene k and tests whether those estimates predict summary statistics in other regions for that gene, while preventing pairs with $r_{jm}^2 \geq \rho_{\text{CV}}$ from being split across training and validation. Unlike chromosome holdout, the size of each holdout unit is determined by the observed LD graph rather than chromosome boundaries; unlike per-gene fold labeling, its label is globally defined.

3 Application to Alzheimer Disease (AD) GWAS

Our default annotation source is the public Nasser et al. ABC prediction table: all element-gene connections with ABC score at least 0.015 across 131 biosamples. The ABC model predicts enhancer-gene regulation from enhancer activity and enhancer-promoter contact; the public table is distributed by the Engreitz Lab and is linked from their resource page. The ABC documentation defines the score as activity times contact, normalized over candidate elements within 5 Mb of the gene promoter, and notes that predictions are same-chromosome CRE-gene links.

For Alzheimer disease (AD), the default run uses the brain-labelled biosamples available in this table: `astrocyte-ENCODE`, `bipolar_neuron_from_iPSC-ENCODE`, and `H1_Derived_Neuronal_Progenitor_Cultured_Cells-Roadmap`. Because the public table does not contain a microglia biosample, the default also includes myeloid proxy contexts: `CD14-positive_monocyte-ENCODE`, `CD14-positive_monocytes-Roadmap`, and `THP-1_macrophage-VanBortle2017`. The full 131-biosample table is downloaded and can be selected by passing `--abc-cell-types all`.

The implementation reads the ABC table in chunks, intersects each CRE interval with GWAS variants, and emits sparse variant-gene-context triples whose values are ABC scores. The regression row universe is not restricted to ABC-overlapping variants. Instead, COMPASS uses all GWAS variants represented in the UK Biobank LD reference panel. Variants outside ABC CREs have all-zero annotation rows but remain in the regression so they contribute to LD tagging and the residual non-ABC-mediated LD-score term. The same reference panel defines global LD components shared by all genes for cross-validation.

LD is represented as chromosome-level sparse blocks rather than a single genome-wide matrix. Raw UK Biobank correlations are converted to r^2, entries with $r^2 < 0.01$ are discarded, and the retained LD values are stored and applied as half-precision sparse tensors during fitting. Other model tensors default to single precision, with a command-line option to compare half-precision model tensors against the default while keeping LD half precision in both runs.

3.1 Implementation status

The repository now downloads the public ABC table into `raw/abc/AllPredictions.AvgHiC.ABC0.015.minus150.ForABCPaperV3.txt.gz` and uses ABC annotations by default in `scripts/run.py`. The row set is filtered to variants present in the UK Biobank LD metadata, and non-CRE rows are retained with zero annotations. For the default ten-fold CV, the implementation constructs chromosome-local connected components at $\rho_{\mathrm{CV}} = 0.01$ directly from cached sparse LD blocks, assigns each component one global fold, and removes every row in that component from the corresponding training loss. Validation scores use only held-out annotated rows linked to genes that also have annotated training components.

3.2 LD-component CV results

In the AD analysis, the $r^2 \geq 0.01$ graph contained 3,354 components across 12.36 million variants. Ten folds were balanced to within one variant, and 104,411 held-out CRE rows were eligible for penalty selection. The held-out error selected the interior value $\lambda = 100$ (mean squared error 2.98059), compared with 2.98074 at $\lambda \in \{300, 1000\}$ and 2.98737 at $\lambda = 30$. Thus, the selected penalty was not at the edge of the tested grid, although its advantage over stronger regularization was modest.

The selected nuclear-norm fit had a zero residual LD-score coefficient and was numerically rank one. A strict rank-one alternating refit initialized from its leading non-negative singular factors reached the same objective within numerical tolerance and had a coefficient correlation of 0.99998 with the nuclear fit. The two parameterizations therefore do not provide independent gene or context resolution in this analysis.

We performed ordered g:Profiler analysis on the top 100 global and context-specific genes from each fit, using all assayed genes as background, and cross-referenced rankings with 967 Agora nominated AD targets. The rank-one structure made context-specific rankings nearly identical. Leading enrichment terms were broad Human Protein Atlas tissue signatures rather than AD- or brain-specific programs, so they are not interpreted as disease mechanisms. At the top 1,000 global genes, 68 nuclear-fit genes and 69 rank-one-fit genes were Agora targets, versus approximately 47 expected from the assayed-gene background (unadjusted hypergeometric $p = 0.00139$ and 0.000877, respectively). The top-100 overlap was 6 targets for each fit and was not significant ($p = 0.33$).

These Agora comparisons are therefore exploratory evidence of broad target-list overlap, not gene-level validation.

3.3 Peripheral-control sensitivity analysis

We repeated the complete analysis after adding three ABC contexts not expected to represent AD-relevant myeloid or neural regulation: white adipose, gastrocnemius muscle, and uterus. This expanded panel contributed 165,905 eligible held-out CRE rows. The ten-fold LD-component score again selected the interior value $\lambda = 100$ (mean squared error 2.91238); the adjacent stronger penalties $\lambda \in \{300, 1000\}$ scored 2.91300. The expanded nuclear solution was again numerically rank one (leading singular value 1.25×10^{-6}; subsequent singular values approximately 10^{-14}), and its rank-one refit had coefficient correlation 0.99998 and relative Frobenius difference 0.0128.

The controls were not down-weighted: their coefficient mass was 8.60% (adipose), 10.36% (muscle), and 10.56% (uterus), comparable to the brain and myeloid contexts (6.85–11.06%). Thus, within the current ABC annotation, LD-score objective, and regularization, the fitted context factor does not distinguish these negative controls from the AD-proximal contexts. The residual non-ABC-mediated coefficient remained zero. Gene rankings and enrichment were correspondingly nearly identical across all contexts. In the expanded assayed-gene background, the top 1,000 global nuclear and rank-one rankings each contained 64 Agora targets (45.6 expected; unadjusted $p = 0.00389$); top-100 overlaps were 7 ($p = 0.17$) and 8 ($p = 0.086$), respectively. These sensitivity results reinforce that the present fit supports only an aggregate component, rather than cell-type-specific AD prioritization.

3.4 Sparse ABC simulation

To distinguish lack of statistical power from annotation identifiability, we simulated direct variant heritability from the same five ABC contexts. Bipolar-neuron and CD14-monocyte annotations received 70% and 30% of total $h^2 = 0.20$, while adipose, muscle, and uterus received zero. In each causal context, only 10% of ABC-positive variants were causal, with direct heritability proportional to their gene-summed ABC scores. Real UK Biobank r^2 propagated these direct effects, and independent noncentral chi-square statistics were drawn at $N_{\text{eff}} = 100{,}000$ and 200,000.

The simulation identifies an annotation-resolution limitation. At $N_{\text{eff}} = 100{,}000$, the nuclear model ranked the two causal contexts above all three controls in two of three replicates; at 200,000, it did so in all three. However, normalized fitted mass remained diffuse at both sample sizes: bipolar and CD14 together received approximately 46–47%, leaving 53–54% on the null controls. The fitted 70:30 split had mean L_1 error approximately 0.54. This occurs despite moderate direct ABC overlap: 36–37% of bipolar-positive variants overlap muscle or uterus annotations, and 41–42% of CD14-positive variants do so. LD propagation and gene summation further reduce context-specific information.

For comparison, we fit the infinitesimal Gaussian reduction in Section 1.1, treating each gene-summed ABC context as an ordinary S-LDSC annotation. The joint regression assigned positive signal to bipolar (implied annotated $h^2 = 0.032$–0.038) and CD14 (0.013–0.015), while joint control estimates were near zero or negative. It therefore better separates the two causal annotations in this sparse simulation, but does not recover their total h^2: the unrestricted residual LD-score coefficient is negative under this intentionally non-infinitesimal, sparse generative mechanism. These signed S-LDSC estimates are a useful comparative ranking baseline, not calibrated heritability estimates under annotation misspecification.

We also ran the official LDSC implementation on the AD statistics, using the same 12.36 million

UK Biobank reference variants, gene-summed ABC scores, and an all-SNP baseline annotation. The observed-scale total estimate was $h^2 = 0.016$ (SE = 0.002). The bipolar-neuron annotation accounted for 0.0028 (SE = 0.0012) of the fitted observed-scale heritability (coefficient 1.21×10^{-7}; $z \approx 2.3$); the CD14-monocyte estimate was 0.0015 (SE = 0.0010), and the three peripheral controls were not individually resolved. The fitted intercept was 1.0155 (SE = 0.0062). This run uses LDSC's native iterative weights and block-jackknife errors, but is not baseline-LD adjusted and uses the analysis $r^2 \geq 0.01$ reference representation; it is therefore a matched comparative analysis rather than a replacement for a conventional HapMap3/baseline-LD S-LDSC study.

We therefore repeated the analysis with the conventional 1000 Genomes European reference, BaselineLD v2.2, HapMap3 regression weights, matching allele frequencies, and overlap-annotation accounting. Using 1,185,548 matched regression SNPs, the observed-scale estimate was $h^2 = 0.0152$ (SE = 0.0022) with intercept 1.0159 (SE = 0.0143). No ABC context had a resolved conditional coefficient. The bipolar-neuron coefficient was 3.56×10^{-8} (SE 5.91×10^{-8}; $z = 0.60$), and the CD14-monocyte coefficient was -1.01×10^{-8} (SE 1.19×10^{-7}; $z = -0.08$); the three peripheral controls were likewise imprecise. Thus the apparent bipolar-neuron signal in the UK Biobank-reference comparison is not robust to standard BaselineLD adjustment. This adjusted result is the appropriate S-LDSC comparison; the earlier matched-reference run remains useful only as a diagnostic of the analysis-specific LD representation.

As an ABC-independent regulatory comparison, we jointly fit binary ATAC-seq, H3K27ac, and H3K4me3 peak annotations from human NeuN neuron, PU1 microglia, Olig2 oligodendrocyte, and LHX2 astrocyte nuclei, again conditional on BaselineLD. PU1 microglial H3K27ac peaks covered 2.72% of SNPs, accounted for 35.4% of fitted h^2 (SE 5.90%), and had 13.0-fold enrichment (SE 2.17; enrichment $p = 2.15 \times 10^{-7}$). Its conditional coefficient was positive (1.58×10^{-7}; SE 5.30×10^{-8}; $z = 2.97$). The binary Glass ABC analysis gave a concordant result: the microglial annotation had 27.1-fold enrichment and a positive conditional coefficient ($z = 3.22$), whereas continuous Glass ABC was unresolved ($z = 1.14$). The agreement between binary ABC support and independent PU1 H3K27ac peaks supports microglial enrichment in AD, but does not establish gene-level mediation.

An uncalibrated hierarchical COMPASS pilot did not recover this specificity. Its microglial H3K27ac coefficient was 1.60×10^{-7} ($z = 1.16$), while neuronal and oligodendrocyte coefficients were larger and had similar uncertainty. After fixing the context direction to the non-negative BaselineLD S-LDSC profile and re-estimating only its scale, the pilot assigned the largest effect to microglia (4.45×10^{-7}; chromosome-jackknife $z = 2.03$). This calibrated result matches the desired cell-type ordering by construction and should be interpreted as a compatibility check, not independent replication. The full LD-component cross-validated fit of the gene-deviation matrix is in progress.

4 Application to Schizophrenia GWAS

The schizophrenia analysis used European-ancestry 2026 summary statistics and the same brain peak panel. BaselineLD-adjusted S-LDSC produced strong and coherent neuronal enrichment. NeuN H3K27ac peaks had 5.71-fold enrichment ($p = 2.14 \times 10^{-23}$) and a positive conditional coefficient (3.08×10^{-7}; SE 4.23×10^{-8}; $z = 7.28$). NeuN H3K4me3 peaks had 10.2-fold enrichment and a positive coefficient ($z = 4.63$). Binary Glass neuronal ABC support was also enriched 10.6-fold, with a positive coefficient ($z = 3.74$); continuous and distal Glass ABC annotations were less well resolved.

Unlike AD, the uncalibrated hierarchical COMPASS H3K27ac pilot independently ranked neurons first. The neuronal coefficient was 3.65×10^{-7} ($z = 2.46$), corresponding to an implied annotated h^2 of 0.216. Astrocytes were also positive ($z = 2.34$), so the result has the expected direction but weaker specificity than S-LDSC. A gene-linked peak-to-expressed-TSS pilot gave the same ordering. Full conditional cross-validation is in progress to test whether gene-specific deviations improve prediction beyond the shared neuronal context effect.